

The Stress of Not Being Able to Perform Dustbathing in Laying Hens

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VESTERGAARD, K. S., E. SKADHAUGE AND L. G. LAWSON. *The stress of not being able to perform dustbathing in laying hens*. *PHYSIOL BEHAV* 62(2) 413-419, 1997. —The stress of not being able to perform dustbathing was studied in small groups of laying hens. The birds were reared and kept for 2.5 years in cages with either sand ("sand birds") or wire floors ("wire birds"), and subsequently deprived of sand (sand birds) or given access to sand (wire birds). Before this change, wire birds had a higher incidence of unilateral wing/leg stretching and stereotypic pecking compared to sand birds; however, there was no difference in corticosterone concentrations. Deprivation of sand in the sand birds resulted in a total absence of dustbathing and in a significant increase in corticosterone concentrations. Although the wire birds dustbathed on the wire before the change of floors, there was a significant increase in the incidence of dustbathing after sand was provided, but no changes in the concentrations of corticosterone. Threats and allopecks decreased in the wire birds after access to sand, whereas no changes were found in the sand birds. Stereotypic pecks on feathers were absent in the sand birds, but were frequent in most of the wire birds both before and after they were given sand. Also, in the wire birds, all feather pecking (stereotypic and nonstereotypic) was positively correlated with the concentration of corticosterone, and this kind of pecking may, thus, be an expression of stress in laying hens. We conclude that the nonperformance of dustbathing behavior is associated with the experience of stress. © 1997 Elsevier Science Inc.

Hens Dustbathing Deprivation Stress behavior Displacement activities Corticosterone Stereotypies

UNDER intensive farming conditions, animals are often deprived of access to stimuli that may be of significance for the performance of normal behavior patterns. This is a potential cause of stress and, in a number of cases, the lack of stimuli and/or space has been associated with behavioral and endocrinological changes indicative of stress and reduced well-being. Fattening pigs kept in crowded and sterile environments provide such an example (25). Also, animals under conditions that severely restrict the performance of behavior, such as sows in crates (8) and bulls in tether on slatted floors (23), provide examples of stress caused by environmental deficiencies. Indeed, such examples of long-lasting deprivation have been suggested to represent major welfare problems in farm animals (10). The main problem with such studies, however, is that the environment inhibits or deprives a wide range of behaviors, so that the significance of performing specific behaviors cannot be revealed.

Deprivation is related to the concept of behavioral needs (18,19), and a crucial question here is whether or not animals have specific needs to perform behavior, even in the case when the result of the behavior under the intensive farming condition has lost its biological significance. Until now, there seems to be no clear-cut analysis to show that the nonperformance of a specific behavior in itself is a stressor to an animal. In such an analysis, a number of indicators of stress (6) should be included to

evaluate how essential it is for the animal to perform the behavior in question.

Dustbathing in fowl provides a good case for the study of deprivation effects and behavioral needs related to a specific behavior because it increases after deprivation (15,38) and because it may appear as a vacuum or sham activity when, for example, performed on a wire floor (36,37). It is a complicated behavior that consists of 15 different elements (22,40), and serves to clean the plumage and keep feathers in good condition (5,14,35). Mature hens on average perform a bout of dustbathing of approximately 0.5 h every other day (38) and, generally, dustbathing is a behavior that takes up substantial energy and time. All this indicates that the performance of dustbathing is significant for the animal, but not necessarily that the nonperformance leads to stress. On the other hand, it has been found recently that low performance of dustbathing behavior was associated with elevated fear in red junglefowl (43).

In the present experiment, we tried to evaluate the potential stress involved in the nonperformance of dustbathing by depriving laying hens of access to sand, and recording the resulting changes in behavioral stress indicators as well as corticosterone concentrations in the blood. We knew from previous experience (38) that transferring hens from litter to wire floor would result in the thwarting of dustbathing, at least for several days. For a

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comparison, we used hens that had been reared on wire floor ("wire birds"). They would provide good "control" birds because they are known to perform vacuum dustbathing on a regular basis (21,24,37,39). We hypothesized that corticosterone concentrations would increase, together with a number of potential behavioral stress indicators, when dustbathing was thwarted in the birds that were deprived of sand after their lifelong access to sand ("sand birds"). For these birds, we also hypothesized that alloprecking or feather pecking (4,42,43) would increase after the change of floors. We decided to record also a number of comfort movements that may appear as displacement movements (2,13,22) and aggressive behavior patterns (11,12) that may occur with a high incidence during frustration and stress. For the wire birds that were given sand after they had been reared and kept on a wire floor, we hypothesized that there would be little change in corticosterone concentrations and behavioral stress indicators, because we knew the birds were regularly dustbathing on the wire floor. However, we did hypothesize that alloprecking would decrease because the sand would provide a strong stimulus for pecking and, consequently, would direct the bird's attention away from feathers (3). By the start of the study, many of the birds had developed stereotypic feather pecking (27) and an additional aim of this study was to examine the relationship between physiological stress as revealed by the corticosterone concentrations and stereotypic pecking and/or feather pecking. We hypothesized that stereotypic pecking was a sign of stress and, consequently, that stereotypic pecking and corticosterone concentrations would be positively correlated.

MATERIALS AND METHODS

Birds and Housing

The birds used were female white leghorn layers ("Shaver Starcross") provided from a commercial dealer. The birds arrived one day old in the laboratory and were from 2 batches with 2 months in between them. At the laboratory, the birds of each batch were allocated to 4 cages (90 × 45 × 45 cm) with 6 birds in each. Two of the cages had sand on the floor and the 2 others had wire floors. They were all provided with 2 small feeding troughs covered with wire, to prevent dustbathing in them, and a tube drinker. The birds in each group were visually isolated from birds in the other groups. A 60-watt "Elstein" heating bulb was placed in each cage until the chicks were 3 weeks old. Light came from a 40-watt lamp above each cage. They were switched on from 0700 to 2000 h (13 h per day). At 7 weeks of age, the birds were transferred to 8 larger cages (approximately 80 × 80 × 80 cm) where they remained until the end of the study, when the birds were approximately 2.5 years old. The cages had either sand or wire on the floor. Two of the wire bird cages and 2 of the sand bird cages were provided with nests with cut straw. A wire floor was placed 3 cm above the nest bottom to prevent the birds from dustbathing in the straw, and we never observed dustbathing in the nests. Otherwise, all cages were similar. They were all provided with drinking nipples and a round trough that was placed in front of the cage. All sides except the front, which was of wire, were solid. There were originally 5 birds in each of these cages; however, because some birds were used in other experiments and because of cannibalism and other causes of death, the number of birds were reduced in most groups. When the study commenced (after approximately 2.5 years) we had 3 groups of 2, 4, and 5 sand birds, respectively, and 4 groups of 2, 3, 4, and 4 wire birds, respectively. Additionally, 11 hens (same strain and line, 1.5 years old) that were not part of the main study, and were continually kept in the same kind of cage environment, served

as controls to examine if there was an effect on corticosterone concentrations of the repetitive blood sampling per se.

The cages of all birds were placed in 2 rooms (3 × 4 m) without windows and with remote control of climate that kept the temperature at 20 ± 1°C. Light came from a 40-watt light bulb placed in front of each cage. It was switched on from 0630 to 1830 h (12 h per day). During observations, there was additional light from 2 40-watt fluorescent light tubes attached to the ceiling. After 3 weeks of observation (see below), the floors were changed so that the sand was removed from the sand birds and replaced by a wire-mesh floor (80 × 80 cm) and the wire floors in the wire bird cages were replaced by sand boxes (80 × 80 cm) filled with sand. Sand was frequently renewed.

Observations

The observations were started 3 weeks prior to the change of floor, when the birds were 32–34 months old, and they were continued during 45 days of exposure to the new floor environment. During the 3 weeks prior to the change of floors, 5 focal observations of 20 min duration were made on each individual, 2 in the morning between 0900 and 1200 h and 2 in the afternoon between 1300 and 1600 h. During the week after the day of change of floors, similar observations were started and carried out over a total of 6 weeks until the end of the study. Observations were balanced according to group, wire vs. sand groups and time of day. The behavior was recorded by direct observation and noted on preformed sheets. The incidence of dustbathing (42) was assessed for all birds in a group during focal observation on any of the group members. We only recorded whether or not dustbathing occurred during an observation and then calculated the rate of occurrence as number of observations with occurrences per total number of observations.

For most behavior patterns, simple rates were calculated for each individual. Comfort behaviors included wing-flapping, yawning, unilateral stretching, bilateral stretching (including intention movements of bilateral stretching = "wings up"), head-flicking (including head-shaking), head-scratching, bill-wiping, body/wing-shaking, tail-shaking, food cleaning (by means of the bill), and feather raising (22). We use the 2 terms head-flicking and head-shaking synonymously (17). For the comfort behavior elements, we hypothesized that keeping the birds on wire would lead to elevated rates. Finally, stereotypic feather pecking was recorded as any repetitive gentle pecking at feathers on penmates ("stereotypic alloprecking") or on the bird itself ("self-pecking"), which also included stereotypic pecks at their own legs). An additional criterion for stereotypic feather pecking was that the pecks should occur in bouts of at least 5 with no more than 1 s between consecutive pecks. The ethogram of social behaviors consisted of aggressive pecks, threats, and alloprecks. All the aggressive pecks, threats, and alloprecks performed and received by the focal individual were recorded and the bird that delivered the behavior toward the focal bird was identified. For the social behavior, we calculated the rate of each element per 20 min delivered by both birds within any pair of birds in a group (43). Because this was done in all observations in which either animal in a pair was the focal animal, the number of observations for which we calculated the rates was based on twice as many observations (i.e., 10 before and 12 after the change of floor) as for the nonsocial behaviors.

From the day of change of floors and until Day 22 thereafter, and additionally on Days 32 and 36, we recorded all dustbathing behavior in the group of 4 sand birds by means of time-lapse video recordings during all the light hours.

Blood Sampling

Blood for the analysis of corticosterone was sampled from all birds by means of heart puncture into chilled heparinized syringes and immediately placed in ice (1) 18 and 12 days before the change of floors and again on Days 0 (the day of change of floors), 12, 18, and 44 after the change of floors. (In addition to the birds that were part of the experiment, the 11 control birds were also sampled on 5 days). Care was taken to minimize the stress of handling prior to sampling. We picked the birds from their group in total darkness 1–2 h after the end of the light period. A person entered the room from the totally dark corridor and seized a bird after 1–2-s orientation by means of a weak electric torch. The bird was immediately carried to a nearby room and sampled. Time from when the bird was seized until the blood was sampled was recorded for all samples and, in almost all cases, it was less than 1 min. After sampling, each bird was placed in a dark box in the dark corridor until all birds were sampled. Then they were returned to their cages in total darkness. When a bird was seized, all the remaining birds in the room were invariably totally calm and made no vocalizations. It was, therefore, our firm impression that no birds were stressed because other birds were removed. Blood samples were centrifuged at 10,000 rpm for 10 min and the plasma stored in a freezer within 2 h after sampling.

The corticosterone assay was carried out exactly as previously described (1).

Statistical Analysis

For each behavior element, the difference between paired observations (before vs. after the change of floors) on each bird within each group were used to test the hypothesis of mean (μ) = 0 with cage as random effect (i.e., accounting for the cage interclass correlation). For behavioral elements where the difference distribution exhibited nonnormality, the logarithmic transformation was applied to minimize variation within the group. To test for group differences, comparison of Group (wire vs. sand birds) was made both for the counts obtained before and for those obtained after the change of floors.

For the corticosterone concentrations, a preliminary test was made to test if the concentration levels differed or were the same in both groups prior to the change of floors (average for Days –18 and –12). Thereafter, the 4 repeated measurements on each bird were subjected to a split-plot analysis. This was done after ensuring that the data satisfied the required correlation structure as tested by the Mauchly's Sphericity test (30). As a control of the model, the residuals were subjected to a test of normality. This test revealed that the residuals were not distributed according to normality, and inspection of the data showed that 2 of the observations of the last measurement (Day 38) were extreme values. The model was re-run without these values and, in this case, the distribution of the residuals did not differ from normality. The result for the group comparisons were the same as when the extreme values were included, although with a slightly stronger p value ($p < 0.01$).

The "proc mixed" procedure of SAS statistical program package (31) with Cage in the random statement was used to analyze the behavioral data, whereas the "proc glm" (30) was used to analyze the corticosterone data. The "proc univariate" (30) was used to perform the test for normality.

In one case, Group differences between groups of sand birds and wire birds were tested by means of the Fisher exact probability test. Correlations were tested by means of the Spearman rank correlation test; see (33) for all these nonparametric tests. Finally, the association between corticosterone concentration and

behavioral stress indicators was examined by means of linear regression analysis.

RESULTS

Dustbathing Behavior

As assessed from the direct observations, dustbathing (Fig. 1) totally disappeared ($p < 0.05$, 2-tailed test) in the sand birds after the change of floor and did not reappear during the 48 days from the change of floor until the end of the experiment. However, a few very short bouts of dustbathing behavior on the wire floor were seen from Day 20 after the change of floors on the video recordings in the group of 4 sand birds that was monitored. As expected, the wire birds performed some dustbathing on the wire floor before the change of floor, and their incidence of dustbathing increased significantly ($p < 0.05$, 2-tailed test) after access to sand. Before the change of floors, the incidence of dustbathing was significantly higher in the sand than in the wire birds ($p < 0.05$, 2-tailed test) whereas, after the change of floors, the wire birds performed significantly more than the sand birds ($p < 0.025$, 2-tailed test).

Comfort and Displacement Behavior

Before the change of floors, the wire birds performed more stretching than the sand birds ($p < 0.05$, 1-tailed test; Fig. 2). After the change of floors, however, there was no tendency for any difference between sand and wire birds. In the sand birds, the counts for total comforts increased as predicted after deprivation of sand, but this result was not statistically significant (Fig. 2). Also, contrary to our hypothesis, none of the comfort elements performed by the sand birds showed a significant increase after deprivation (i.e., after change of floors). Similarly, in the wire birds, there was no tendency of any decrease (or increase) after the change of floors.

Stereotypic Feather Pecking and AlloPecking

No stereotypic allopecks or self-pecks were seen in the sand birds. In the wire birds, however, they were frequent and there were no significant differences from before to after the



FIG. 1. Incidence of dustbathing (means) in sand and wire birds before and after change of floors. * $p < 0.05$, significant changes within sand or wire birds after the change of floors; $p < 0.025$, sand birds, in comparison with wire birds, had a significantly higher incidence of dustbathing before the change of floors and wire birds had a significantly higher incidence of dustbathing than the sand birds after the change of floors.

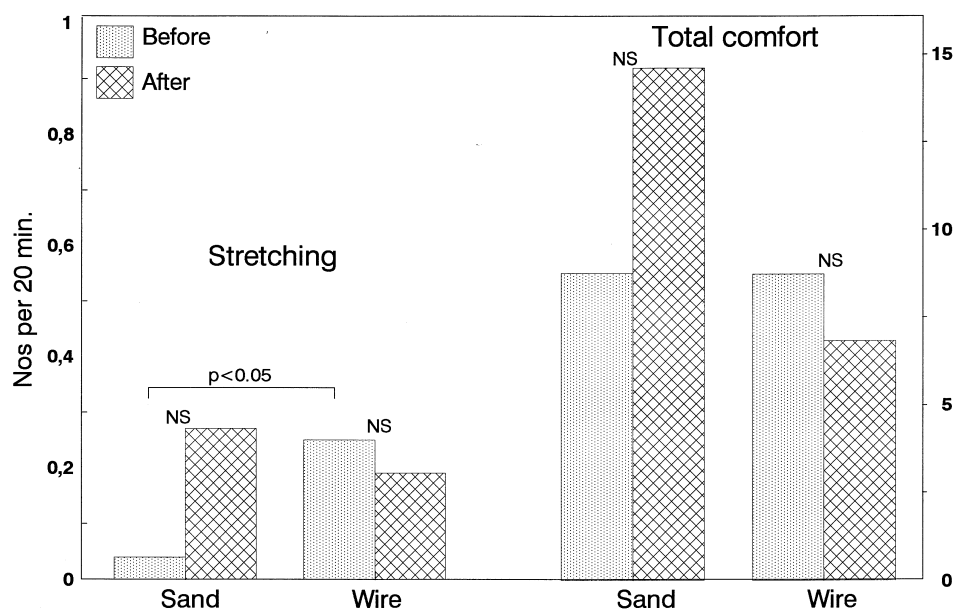


FIG. 2. Rates of unilateral wing/leg stretching (stretching) and "total comforts" in sand and wire birds before and after the change of floors. Change of floor had no significant influence on the incidences of comfort behavior patterns although, for total comforts, the results were in the predicted direction. However, before the change of floors, there was significantly more stretching in the wire than in the sand birds ($p < 0.05$). Total comforts includes all the comfort patterns mentioned in the introduction.

change of floors (Fig. 3). In the wire birds, also, the average numbers of all stereotypic pecks (stereotypic allopecks + stereotypic selfpecks) were almost identical before and after the

change of floors (see Fig. 3). Altogether, 10 of the 13 wire birds performed stereotypic pecking. Eight birds performed stereotypic allopecking and 5 birds performed stereotypic self

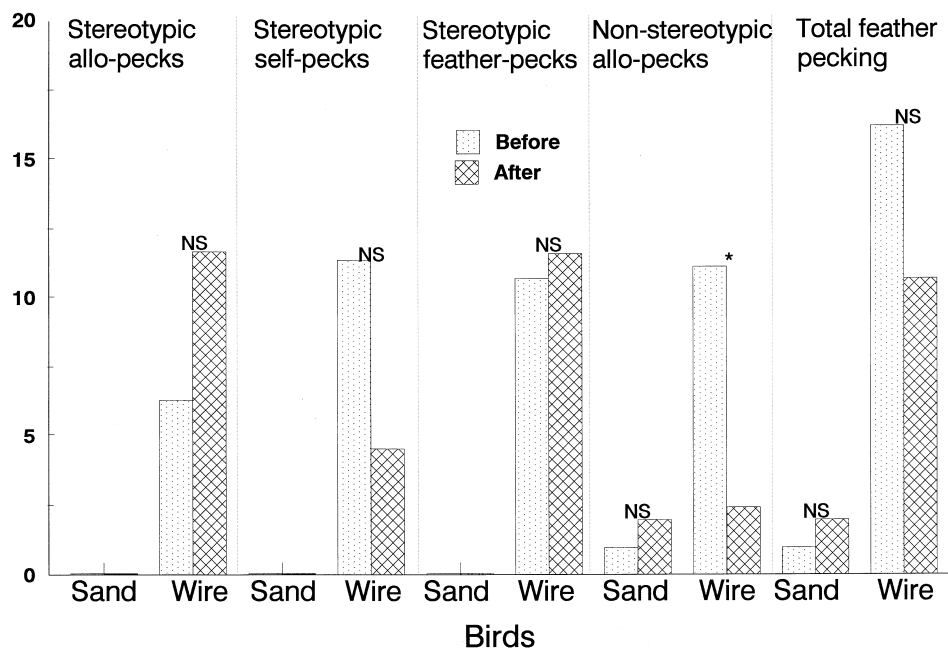


FIG. 3. Rates of stereotypic feather pecking and of allopecks (nonstereotypic allopecking) in sand and wire birds before and after the change of floors. Stereotypic feather pecking was the sum of stereotypic allopecks and stereotypic selfpecks, and total feather pecking was the sum of stereotypic feather pecks and nonstereotypic allopecks. * $p < 0.05$, significant reduction in wire birds from before to after the change of floors; NS, nonsignificant change. There was a significantly higher rate of allopecking in the wire birds before the change of floors ($p < 0.025$).

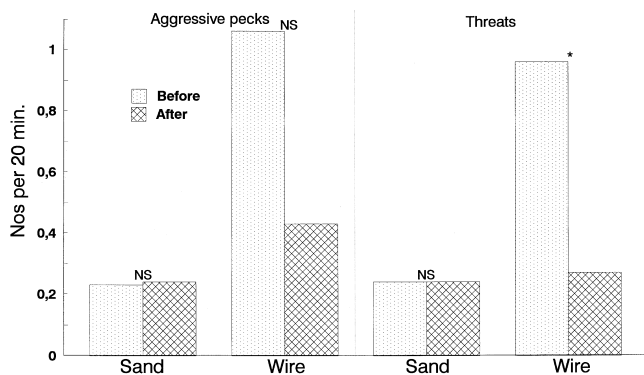


FIG. 4. Rates of aggressive pecks and threats in the pair-relationships of sand and wire birds before and after change of floors. * $p < 0.05$, significant change from before to after change of floors in the rate of threats in wire birds.

pecks. Two of these birds exclusively performed stereotypic self pecking.

The number of birds that showed stereotypic feather pecking was significantly higher among wire birds than among the sand birds (there were none among the sand birds) as revealed by the Fisher exact probability test ($p < 0.001$ for all stereotypic feather pecking and $p < 0.01$ for both stereotypic allopecking and stereotypic self pecking).

Changes in Social Behavior

In the sand birds, there were no changes from before to after the transfer to wire in the rates of social behavior elements within pairs (pair relationships; Fig. 4). In the wire birds, however, there was a significant reduction in threats after the change of floor (i.e., after access to sand) ($p < 0.05$; Fig. 4). The pair-relationship data for allopecking failed to reveal significant differences due to change of floors or to reveal significant differences between sand and wire birds.

When individual rates were examined, there was no significant increase in nonstereotypic allopecking in the sand birds after deprivation (Fig. 3). On the other hand, the rate of nonstereotypic allopecks in wire birds became reduced after access to sand ($p < 0.05$). Before the change of environment, the sand birds made fewer nonstereotypic allopecks than the wire birds ($p < 0.025$; Fig. 3) whereas, after the change of floors, there was no significant difference between sand and wire birds.

Changes in Corticosterone Concentrations

Before the change of floors (Day -18 and Day -12), there was no difference between sand and wire birds in the median corticosterone concentrations and, also, there were no differences between Day -18 and Day -12 (Fig. 5). After the change of floors, however, the corticosterone concentration of sand birds became significantly higher than that of the wire birds ($p < 0.05$; Fig. 5).

In the control group of 11 hens that were given no change of environment, blood sampling was started 5 months later than in the main experiment, on 5 different days with intervals of 7, 14, 7, and 17 days, respectively. In these birds, no consistent or significant changes were found over the consecutive days of blood sampling. The overall mean of the median value for each of the birds was 1.80 (1.35–2.94) ng per ml, which was essentially the same as that of the wire birds before change of floors (1.79 ng

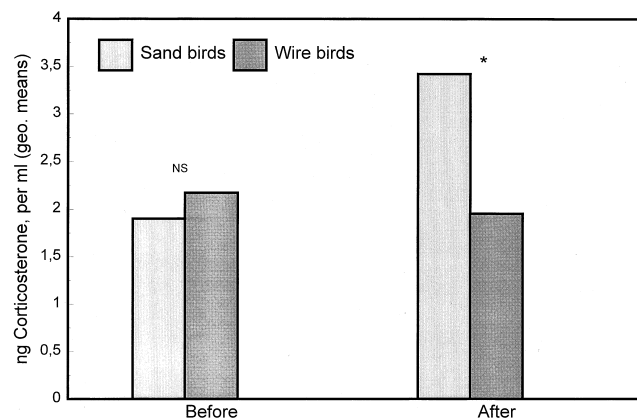


FIG. 5. Changes in the levels of corticosterone in sand and wire birds from before the change of floors (mean values for Days -18 and -12 combined) to after the change of floor (Days 0, 8, 15, and 38 combined). After the change of floors, there was a significantly higher corticosterone concentration in sand than in wire birds (* $p < 0.05$).

per ml). We conclude that the repetitive blood sampling had no influence on the levels of corticosterone concentration.

In the wire birds, the median values of the 6 corticosterone concentrations of each bird were positively correlated with "all stereotypic pecking" performed ($r_s = 0.48$; $p < 0.05$, 1-tailed) and with the combined values for all stereotypic feather pecking and allopecking at feathers (i.e., total feather pecking performed) ($r_s = 0.67$; $p < 0.01$, 1-tailed). A linear regression analysis gave a similar result for all stereotypic pecking and allopecking (Fig. 6; R-square = 0.34, slope = 11.13, $p < 0.036$). In the sand birds (which had no pecking stereotypies), there was a significant negative correlation after the change of floors between median values of corticosterone and head-flicking (Spearman rank correlation test, $n = 11$, $p < 0.027$, 2-tailed). No correlation between head-flicking and plasma concentration of corticosterone was found in the wire birds.

DISCUSSION

The main result of this study is that the birds reared and continually kept with access to sand after deprivation from sand became unable to dustbathe, and that this was associated with ele-

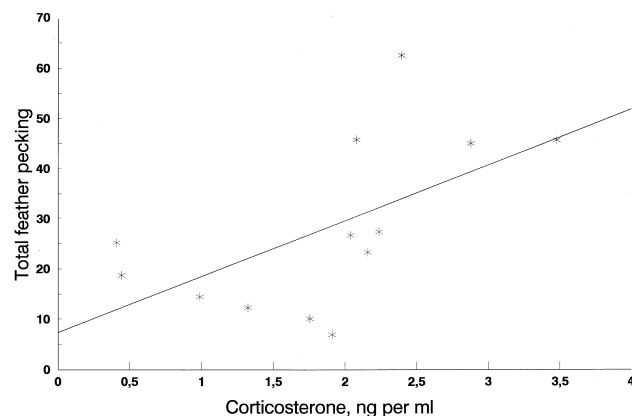


FIG. 6. The association between total feather pecking (all stereotypic pecking + allopecking) and corticosterone levels in wire birds (linear regression, $p < 0.036$).

vated plasma levels of corticosterone and changes in behavior indicative of stress. On the other hand, the absence of a physiological response in the wire birds may be taken as evidence that imposing a major environmental change does not necessarily result in physiological evidence of stress. Additionally, the results from the control birds that remained in the same environment indicate that blood sampling alone was not the cause of the physiological response observed in the sand birds. Thus, this study has revealed a clear case of stress in an animal when an appropriate stimulus for the release of a specific behavior pattern (i.e., dustbathing) is missing.

Dustbathing in fowls reacts to deprivation in such a way that an elevated performance proportional to the time of deprivation is seen when the stimulus (sand) is given again after the end of deprivation [e.g., (38)]. This kind of rebound effect has been discussed in detail by Nicol (26), and it was suggested that the effect was due either to a gradual increase in internal causal (motivational) factors during deprivation or to the novelty of the environment that was presented after deprivation. In the present study, the sand would be the novel stimulus that might have released the novelty response (i.e., the increased performance of dustbathing in the wire birds after presentation of sand). However, novelty aspects of a stimulus are usually short-lived and, for example, in pigs a response to a novel stimulus may fall steeply over the first 5 min (45). In the present study, our data were collected over 7 weeks, starting some days after the change of floors and, accordingly, it seems irrelevant to emphasize novelty in the explanation of our dustbathing data. Potentially long deprivation may lead to frustration and, consequently, stress (10) and, in this study, it is most likely that a gradual increase in motivation in the sand birds after the change of floors was the cause of their elevated corticosterone concentration. Other data point in the same direction. For example, a low performance of dustbathing in red junglefowl has been found associated with increased fear (43), so it may be that the nonperformance of dustbathing (also in the presence of sand) may have a number of behavioral consequences in addition to the ones that have been revealed in the present study. It has also been shown that the infusion of corticosterone in laying hens, in addition to elevating the plasma concentration of corticosterone and the heterophilic/lymphocyte ratio, also increased their fear as measured by tonic immobility (20). Thus, there may be a link between the lack of performance of dustbathing on the one hand and fear and physiological indicators of stress on the other hand.

As mentioned above, the results of the behavior observations may be taken as a stress response associated with being kept on wire. This interpretation is, first of all, based on the result that only wire birds showed stereotypic pecking and wire birds also showed a higher level of stretching than the sand birds (before the change of floors). The development of stereotypic behavior in fowl has previously been found associated with thwarting and frustration (12) and, similarly, increased stretching (unilateral wing/leg stretching) has previously been found after litter-reared birds were transferred to battery cages (i.e., an environment with wire floor and no litter) (2). We see no reason why the wire environment should influence stretching directly and the increased incidence may, therefore, be interpreted as a displacement behavior indicative of frustration and stress. For the remaining comfort or displacement movements, however, we failed to find effects of the treatments. In addition to the results for stereotypic behavior and stretching, the reduced aggression

in the wire birds after the change of floors also supports the interpretation that their stress was higher when kept on wire than when kept with access to sand [see (11) and (16)].

In the present study, a distinction between allopreening pecks, which refer to social preening (or social grooming), and feather pecking could not be made. To do that, it is crucial to record where the victims were pecked (43), but this was unfortunately not made. However, the allopecking in this study was probably mainly feather pecking (i.e., pecks that may damage the feathers) (40). Although the rate of allopecking (nonstereotypic allopecks) decreased significantly in wire birds after they were given access to sand, there was no increase in the sand birds after deprivation (i.e., after the change of floors). Thus, there was no consistent change with the presence of sand. However, in a study in which young hens were reared on either litter (wood shavings) or on wire floors and then at 18 weeks of age transferred to laying pens with litter (4), less allopecking was found in the litter-reared birds, which is a result similar to that of the present study. The most likely interpretation is that wire-reared birds run a risk of becoming imprinted on feathers for pecking in a dustbathing context (43,44), so that the pecking at feathers continues to some extent after the transfer to litter (or sand).

The decreased incidence of allopecking in the wire birds after they were given access to sand may have appeared because sand is a stronger stimulus for pecking than feathers [e.g., (29,41)] and/or because the stress of the wire birds was somewhat alleviated because the birds were dustbathing more frequently in the sand than previously on the wire. The association between total feather pecking (and total stereotyped pecking) and corticosterone concentrations in the wire birds supports this conclusion. However, the lack of a decrease in corticosterone levels in the wire birds after they were given access to sand does not support that conclusion. This lack of response may be explained from the results of stress studies in old laying hens in battery cages, indicating that the ACTH-adrenal cortical system becomes exhausted as a result of long exposure to a stressful environment (7,9). Exactly the same may have been the case in our wire birds, which, until the change of floors, had low corticosterone levels that were similar to those of the sand birds. The elevated aggression and allopecking before the change of floor and also the pecking stereotypies may, nonetheless, indicate that the wire floor was stressful for these birds.

The correlation between the amount of stereotypic pecking and levels of corticosterone is a new and interesting finding. No such correlation has been found between abnormal or stereotypic behaviour in cattle and the adrenal cortex response (23,28), nor has it been found in studies on stereotypic behavior in tethered sows (34) or, as far as we know, in other farm animals. However in a recent paper, a negative correlation was found between plasma corticosterone and time occupied in stereotypic behavior in restricted-fed broiler breeders (32). This response, however, may be very different from that of our hens because it was clearly related to feeding whereas, in our hens, the stereotypic behavior was related to the absence of sand for dustbathing.

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